Mathematics for Intelligent Systems 3

23MAT204

Linear Programming (LP)

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Disclaimer

These slides may contain unintentional errors.
I would greatly appreciate it if you could point them out.
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References

Main references:

- *Convex Optimization*, Stephen Boyd and Lieven Vandenberghe
  https://web.stanford.edu/~boyd/cvxbook/bv_cvxbook.pdf
- *Operations Research*, Taha, H. A. : An Introduction, 10th ed., Pearson (2017)

General Form of Linear Programs

A linear program can take many forms:

- **Objective:** Maximize or Minimize
- **Constraints:** Equalities or inequalities
- **Variables:** Restricted ($x_j \geq 0$) or unrestricted ($x_j \gtrless 0$)

General form:

$$\text{Maximize/Minimize } z = c_0 + c_1 x_1 + \cdots + c_n x_n$$
$$\text{subject to } a_{i1} x_1 + \cdots + a_{in} x_n \ (\leq, =, \geq) \ b_i, \quad i = 1, \ldots, m$$

Definition: Standard Form

A linear program is in **standard form** if:

- It is a maximization problem,
- Constraints are equalities only, and
- All variables are nonnegative.

Matrix form:

$$\text{Maximize } z = c^T x \quad \text{subject to } Ax = b,\ x \geq 0.$$

Definition: Canonical Form

A linear program is in **canonical form** if:

$$\begin{aligned} &\text{Maximize } z = c^T x, \\ &\text{subject to } Ax \leq b, \\ &x \geq 0. \end{aligned}$$

Conversion:

- Canonical to Standard: $Ax \leq b \implies Ax + Is = b, \quad s \geq 0.$

Slack Variables

For a $\leq$ constraint:

$$a_1x_1 + a_2x_2 \leq b \quad \Rightarrow \quad a_1x_1 + a_2x_2 + s = b, \quad s \geq 0.$$

Example:

$$x_1 + x_2 \leq 5 \quad \Rightarrow \quad x_1 + x_2 + s = 5, \quad s \geq 0.$$

Surplus Variables

For a $\geq$ constraint:

$$a_1x_1 + a_2x_2 \geq b \quad \Rightarrow \quad a_1x_1 + a_2x_2 - s = b, \quad s \geq 0.$$

Example:

$$2x_1 + 3x_2 \geq 8 \quad \Rightarrow \quad 2x_1 + 3x_2 - s = 8, \quad s \geq 0.$$

Artificial Variables

Artificial variables are introduced to aid in finding an initial basic feasible solution when none is obvious:

- Added temporarily to equations,
- Penalized in the objective function (Big M or Two-Phase methods).

Example:

$$x_1 + x_2 = 4 \quad \Rightarrow \quad x_1 + x_2 + a = 4, \quad a \geq 0.$$

Feasible Region

The **feasible region** is the set of all points $(x_1, x_2, \ldots, x_n)$ satisfying all constraints:

- It is a convex polyhedron,
- Every feasible solution lies inside or on its boundary.

Example:

$$x_1 + x_2 \leq 5, \quad x_1, x_2 \geq 0,$$

which forms the triangular region bounded by the axes and the line $x_1 + x_2 = 5$.

Basic Variables

- Correspond to columns of the identity matrix in the constraint equations.
- Have coefficient 1 in one equation and 0 in all others.
- Their values are obtained directly from the right-hand side constants.

Example:

$$x_1 + x_2 + s_1 = 6, \quad x_2 + s_2 = 4.$$

Here, s_1 and s_2 are basic variables.

Non-Basic Variables

- Variables not in the basis.
- Assigned value 0 in basic feasible solutions.
- Can change during iteration.

Example:

x_1, x_2 are non-basic when s_1, s_2 are basic.

Basic Solution

Given a system of m simultaneous linear equations with $n > m$ variables:

$$Ax = b,$$

where A is an $m \times n$ matrix with $\mathrm{rank}(A) = m$.

- Let B be any $m \times m$ non-singular submatrix of A formed by reordering m linearly independent columns of A.
- Set the $n - m$ variables not associated with columns of B to zero.
- Solve the resulting system:

$$x_B = B^{-1}b$$

  to get a **basic solution**.
- The m variables in x_B are called **basic variables**.
- B is called the **basis matrix**, and its columns are **basis vectors**.

Basic Feasible Solution

A **basic solution** x_B is called **basic feasible** if

$$x_B \geq 0.$$

This means the solution satisfies the non-negativity constraints in addition to the system of equations.

Summary:

- Basic feasible solutions represent vertices (corner points) of the feasible region.

Optimal Solutions

- The **optimal solution** lies at a vertex (corner point) of the feasible region.
- Multiple vertices can yield the same optimal value, forming an **optimal region**.
- Linear programs always achieve the optimum on the boundary (if bounded and feasible).

Cost Vector

Consider the linear program:

$$\begin{aligned} \max \quad & Z = c^T x \\ \text{subject to} \quad & Ax = b, \\ & x \geq 0. \end{aligned}$$

If x_B is a basic feasible solution, then the **cost vector** corresponding to the basis B is

$$c_B = (c_{B1}, c_{B2}, \ldots, c_{Bm}),$$

which are the coefficients of the basic variables in the objective function. The cost vector is used in the Simplex method to evaluate the optimality of the current basic feasible solution.

Example 1: General to Standard Form

Problem:

$$\text{Maximize } z = 2x_1 - x_2$$

subject to

$$\begin{aligned} x_1 + x_2 &\leq 2, \\ 3x_1 + x_2 &\leq 4, \\ x_1 \geq 0,\ x_2 &\geq 0. \end{aligned}$$

Convert to standard form
with slack/surplus/artificial variables.

Example 1: Standard Form Representation

$$\text{Maximize } z' = 2x_1 - x_2 + 0s_1 + 0s_2$$

$$x_1 + x_2 + s_1 = 2,$$

$$3x_1 + x_2 + s_2 = 4,$$

$$x_1, x_2, s_1, s_2 \geq 0.$$

Decision variables: x_1, x_2, s_1, s_2. **Note:** s_1, s_2 (slack).

Example : General to Standard Form

Problem:

$$\text{Maximize } z = 7x_1 - 5x_2 + 3x_3$$

subject to

$$\begin{aligned} x_1 + 2x_2 &\geq 2, \\ 3x_2 + x_3 &\leq 4, \\ 2x_1 + x_2 + 4x_3 &\leq 4, \\ x_1 \geq 0,\ x_2 \geq 0,\ x_3 &\geq 0. \end{aligned}$$

Convert to standard form
with slack/surplus/artificial variables.

Example: Standard Form Representation

$$\text{Maximize } z' = 7x_1 - 5x_2 + 3x_3 + 0s_1 + 0s_2 + 0s_3$$

$$x_1 + 2x_2 - s_1 = 2,$$

$$3x_2 + x_3 + s_2 = 4,$$

$$2x_1 + x_2 + 4x_3 + s_3 = 4,$$

$$x_1, x_2, s_1, s_2, s_3 \geq 0.$$

Decision variables: x_1, x_2, s_1, s_2, s_3. **Note:** s_1 (surplus), s_2, s_3 (slack).

Example: Conversion to Standard Form

Original problem:

$$\text{Minimize } z = 2x_1 - x_2, \quad x_1 + x_2 \geq 2, \quad x_1 \gtrless 0, \quad x_2 \geq 0.$$

Converted to standard form (introducing slack/surplus and splitting variables):

$$\begin{aligned}
&\text{Maximize } z' = -2x_1^+ + 2x_1^- + x_2 + 0x_3 + 0x_4, \\
&\text{subject to} \\
&x_1^+ - x_1^- + x_2 - x_3 = 2, \\
&3x_1^+ - 3x_1^- + 2x_2 + x_4 = 4, \\
&x_1^+ - x_1^- + 2x_2 = 3, \\
&x_j \geq 0, \quad j = 1^+, 1^-, 2, 3.
\end{aligned}$$

Example : General to Standard Form

Problem:

$$\text{Minimize } z = 2x_1 - x_2$$

subject to

$$\begin{aligned} x_1 + x_2 &\geq 2, \\ 3x_1 + 2x_2 &\leq 4, \\ x_1 + 2x_2 &= 3, \\ x_1 \gtrless 0,\ x_2 &\geq 0. \end{aligned}$$

Convert to standard form:
with slack/surplus/artificial variables.

Example : Standard Form Representation

$$\text{Maximize } z' = -2x_1^+ + 2x_1^- + x_2 + 0x_3 + 0x_4$$

$$\begin{aligned} x_1^+ - x_1^- + x_2 - x_3 &= 2, \\ 3x_1^+ - 3x_1^- + 2x_2 + x_4 &= 4, \\ x_1^+ - x_1^- + 2x_2 &= 3, \\ x_1^+, x_1^-, x_2, x_3, x_4 &\geq 0. \end{aligned}$$

Decision variables: $x_1^+, x_1^-, x_2, x_3, x_4$. **Note:** x_3 (surplus), x_4 (slack).

Example : Solving LP by trial and error method

Problem:

$$\text{Maximize } z = x_1 + 3x_2 + 3x_3$$

subject to

$$\begin{aligned} x_1 + 2x_2 + 3x_3 &= 4, \\ 2x_1 + 3x_2 + 5x_3 &= 7, \\ x_1 \geq 0,\ x_2 \geq 0,\ x_3 &\geq 0. \end{aligned}$$

Basic solutions

S.No. of the basic solution	*Basic variables*	*Non-basic variables*	*Values of the basic variables given by the constraint equations*	*Value of the objective function*	*Is the solution feasible? (are all $x_j \geq 0$?)*	*Is the solution non-degenerate (are all basic variables > 0?)*	*Is the solution feasible and optimal?*
1	x_1, x_2	x_3	$x_1 + 2x_2 = 4$				
			$2x_1 + 3x_2 = 7$	5	Yes	Yes	Yes
			$\therefore \quad x_1 = 2, x_2 = 1$				
2	x_1, x_3	x_2	$x_1 + 3x_3 = 4$				
			$2x_1 + 5x_3 = 7$	4	Yes	Yes	No
			$\therefore \quad x_1 = 1, x_3 = 1$				
3	x_2, x_3	x_1	$2x_2 + 3x_3 = 4$				
			$3x_2 + 5x_3 = 7$	—	No	No	No
			$\therefore \quad x_2 = -1, x_3 = 2$				

Example : Solving LP by trial and error method

Problem:

$$\text{Maximize } z = 3x_1 + 4x_2$$

subject to

$$\begin{aligned} x_1 + x_2 &\leq 450, \\ 2x_1 + x_2 &\leq 600, \\ x_1 \geq 0,\ x_2 &\geq 0. \end{aligned}$$

Example: Standard Form Representation

$$\text{Maximize } z' = 3x_1 + 4x_2 + 0s_1 + 0s_2$$

$$x_1 + x_2 + s_1 = 450,$$

$$2x_1 + x_2 + s_2 = 600,$$

$$x_1, x_2, s_1, s_2 \geq 0.$$

Decision variables: x_1, x_2, s_1, s_2. **Note:** s_1, s_2 (slack).

Basic solutions

S.No. of the basic solution	*Basic variables*	*Non-basic variables*	*Values of the basic variables given by the constraint equations*	*Value of the objective function*	*Is the solution feasible? (are all $x_j \geq 0$?)*	*Is the solution non-degenerate (are all basic variables > 0?)*	*Is the solution feasible and optimal?*
1	x_1, x_2	s_1, s_2	$x_1 + x_2 = 450$ $2x_1 + x_2 = 600$ $\therefore x_1 = 150, x_2 = 300$	1,650	Yes	Yes	No
2	x_1, s_1	x_2, s_2	$x_1 + s_1 = 450$ $2x_1 = 600$ $\therefore x_1 = 300, s_1 = 150$	900	Yes	Yes	No
3	x_1, s_2	x_2, s_1	$x_1 = 450$ $2x_1 + s_2 = 600$ $\therefore x_1 = 450, s_2 = -300$	—	No	No	No
4	x_2, s_1	x_1, s_2	$x_2 + s_1 = 450$ $x_2 = 600$ $\therefore x_1 = -150$	—	No	No	No
5	x_2, s_2	x_1, s_1	$x_2 = 450$ $x_2 + s_2 = 600$ $\therefore s_2 = 150$	1,800	Yes	Yes	Yes
6	s_1, s_2	x_1, x_2	$s_1 = 450, s_2 = 600$	0	Yes	Yes	No